BACKEND OF FLIGHT BOOKING SYSTEM

MINI PROJECT REPORT

Submitted in partial fulfilment of the requirement for the award of the degree of

Bachelor of Technology

in

Computer Science & Engineering

Submitted to

DR. A.P.J. ABDUL KALAM TECHNICAL UNIVERSITY, UTTAR PRADESH, LUCKNOW

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CERTIFICATE

This is to certify that the Mini Project Report entitled "Backend of flight booking system" is a bonafide work carried out by Vikas (Roll No. 2301530100149) in partial fulfilment of the requirements for the award of the degree of Bachelor of Technology in Computer Science & Engineering during the academic year 2026–2027, 7th Semester (Section 7/A), at the Department of Computer Science & Engineering, Lloyd Institute of Engineering & Technology.

The work embodied in this report has not been submitted elsewhere for the award of any other degree or diploma. To the best of our knowledge, the report fulfils the prescribed structure and formatting requirements of the Mini Project guidelines.

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DECLARATION

I hereby declare that the Mini Project Report entitled "Backend of flight booking system" submitted to the Department of Computer Science & Engineering, Lloyd Institute of Engineering & Technology, in partial fulfilment of the requirements for the degree of Bachelor of Technology in Computer Science & Engineering, is a record of original work carried out by me under the guidance of Ms. Deepika Arora, Assistant Professor.

I further declare that this report has not been copied from any other source, and that all references, architectural diagrams, schemas, and source code taken from other sources or created during development have been duly acknowledged and documented.

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ABSTRACT

The civil aviation industry demands high-throughput, fault-tolerant transaction processing to govern flight schedules, multi-cabin seat inventories, dynamic airfares, ancillary meal reservations, and electronic ticketing. Traditional monolithic Passenger Service Systems (PSS) suffer from tight operational coupling, single points of failure, and horizontal scaling bottlenecks during seasonal booking surges. This project introduces the Backend of flight booking system, an enterprise-grade distributed microservices platform built using Java 17, Spring Boot, Spring Cloud OpenFeign, Apache Kafka, PostgreSQL, and Razorpay Gateway.

The architecture decomposes commercial aviation operations into nine autonomous microservices: user-service, airline-core-service, flight-ops-service, location-service, pricing-service, Seat-Service, ancillary-service, booking-service, and payment-service, supported by a shared common-lib module. To guarantee data sovereignty, each service operates an independent PostgreSQL database. Inter-service workflows leverage synchronous OpenFeign clients for low-latency queries alongside asynchronous Apache Kafka event messaging (BookingConfirmedEvent, PaymentCompletedEvent, PaymentFailedEvent) to achieve eventual consistency and non-blocking payment reconciliation. Stateless security is enforced via JJWT HMAC-SHA256 bearer tokens and multi-tenant header isolation. Empirical test suites executed across 15 functional scenarios demonstrate sub-5ms cached airport lookups, safe multi-seat reservations, and automated prevention of double-spend attacks via Razorpay cryptographic signature verifications.

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LIST OF TABLES

(System specification and verification tables are documented in Chapter 3 onwards as per the project roadmap)

LIST OF ABBREVIATIONS

Abbreviation	Full Form
API	Application Programming Interface
BCrypt	Adaptive Password Hashing Algorithm
CRUD	Create, Read, Update, Delete
DAO	Data Access Object
DFD	Data Flow Diagram
DTO	Data Transfer Object
ER	Entity Relationship
GDS	Global Distribution System
HMAC	Hash-Based Message Authentication Code
HTTP	Hypertext Transfer Protocol
IATA	International Air Transport Association
ICAO	International Civil Aviation Organization
JAR	Java Archive
JDBC	Java Database Connectivity
JSON	JavaScript Object Notation
JWT	JSON Web Token
ORM	Object-Relational Mapping
PNR	Passenger Name Record
POJO	Plain Old Java Object
PSS	Passenger Service System
RDBMS	Relational Database Management System
REST	Representational State Transfer
SDK	Software Development Kit
SHA	Secure Hash Algorithm
SQL	Structured Query Language
UML	Unified Modeling Language

CHAPTER 1: INTRODUCTION

1.1 Background

Commercial civil aviation represents one of the most operationally intricate and latency-sensitive domains in global enterprise computing. On any given calendar day, commercial airlines orchestrate hundreds of thousands of individual flight departures across international borders, coordinating passenger itineraries, cabin class seat allocations, customized baggage allowances, flight catering provisions, and high-value digital monetary settlements. Modern air travelers demand instantaneous access to real-time seat availability, dynamic airfare calculation, zero-latency payment checkouts, and immediate electronic ticket issuance.

Historically, airline reservation software relied upon monolithic Passenger Service Systems (PSS) and legacy mainframe Global Distribution Systems (GDS) pioneered in the mid-twentieth century. Although dependable for primitive centralized ledger updates, these monoliths suffer from rigid data schemas, operational coupling, and horizontal scaling bottlenecks under peak seasonal traffic (such as holiday flash sales or unforeseen route cancellations). A single software defect in a secondary module—such as baggage metadata or ancillary meal logs—frequently cascaded to destabilize the core flight reservation engine.

With the advent of cloud-native computing, enterprise software has transitioned toward the Microservices Architecture. By decomposing monolithic domains into decoupled, business-aligned services with independent persistence stores, platforms achieve horizontal scalability, fault isolation, independent CI/CD release cycles, and real-time asynchronous reconciliation.

1.2 Problem Statement

Engineering a high-performance distributed backend for an airline booking ecosystem presents several critical technical and operational challenges:

- **Cascading Failures and Single Points of Failure (SPOF):** In traditional monolithic backends, runtime memory exhaustion or database lock deadlocks in auxiliary modules immediately compromise mission-critical customer bookings.
- **Distributed Inventory Consistency:** Partitioning databases across independent microservices creates the hazard of stale flight instance availability, phantom bookings, or double-selling of physical aircraft seats.
- **Operational Lifecycle Complexity:** Airlines require strict coordination across diverse domains: governmental IATA/ICAO code governance, fleet specs (cabin layout, cruising speed, range), dynamic baggage rules, recurring flight schedules, and live flight seat instances.
- **Payment Security & Fraud Prevention:** Financial checkout flows demand tamper-proof order generation and cryptographic signature verification (such as Razorpay HMAC-SHA256) without introducing blocking latency into passenger reservation threads.

1.3 Objectives

The principal objectives of the Backend of flight booking system are:

- **Domain-Driven Microservice Decomposition:** Architect and implement nine discrete Spring Boot microservices (user-service, airline-core-service, flight-ops-service, location-service, pricing-service, Seat-Service, ancillary-service, booking-service, payment-service) coupled via a common domain library (common-lib).
- **Database-per-Service Pattern:** Deploy nine independent PostgreSQL databases on isolated schemas, eliminating cross-service SQL joins.
- **Hybrid Synchronous and Asynchronous Integration:** Integrate services synchronously using Spring Cloud OpenFeign for real-time queries and asynchronously via Apache Kafka message brokers for event-driven transaction completion.
- **Stateless Authentication & Multi-Tenancy:** Implement stateless security via JJWT (HMAC-SHA256) bearer tokens and enforce tenant isolation through custom forwarded headers (X-User-Id).
- **Cryptographic Payment Verification:** Integrate Razorpay Java SDK for order creation and HMAC webhook validation, utilizing Kafka to transition bookings from PENDING to CONFIRMED.
- **Sub-5ms Navigation Caching:** Implement in-memory cache abstraction (ConcurrentMapCacheManager) on high-frequency airport lookups.

1.4 Scope of the Project

The functional and technical scope encompasses:

- **Identity & Access Management:** Registration, BCrypt password hashing, and role-based access control (ROLE_USER, ROLE_AIRLINE_OWNER, ROLE_SYSTEM_ADMIN).
- **Fleet & Route Administration:** Airline registration, admin approvals/suspensions, aircraft fleet configuration, and international airport route scheduling.
- **Seat Map & Availability Engine:** Cabin class definitions, multi-column seat map grid generation, and real-time seat instance availability tracking.
- **Pricing & Ancillary Add-ons:** Base fare structures, baggage policy rules, dietary meal catalogs, and total surcharge calculations.
- **Booking & Payment Lifecycle:** Multi-passenger booking creation, PNR generation, payment link generation, cryptographic verification, and e-ticket generation.

1.5 Organisation of the Report

The report is organized as follows: Chapter 1 provides the foundational context, problem statement, objectives, and project scope. Subsequent chapters detail literature review, system analysis and design, microservices implementation, testing strategies, empirical results, and future enhancements.